

Software Requirements Specification (SRS)

Municipal Infrastructure Damage Reporting App

Problem Statement #29 | Domain: Smart Cities, Transport & Logistics

1. Introduction

1.1 Purpose

This document specifies the functional and nonfunctional requirements for the Municipal Infrastructure Damage Reporting App, a citizen-facing portal that allows residents to report road potholes, broken streetlights, and water pipeline leaks using geotagged photos. The system automatically extracts location data and dispatches tickets to the relevant municipal engineering department.

1.2 Scope

The system covers citizen-side report submission and status tracking, automated GPS extraction and GIS-based location pinning, automated ticket routing to municipal departments, and engineer-side ticket review and resolution workflows. Payment processing, citizen account billing, and non-municipal (private property) damage reports are out of scope.

1.3 Intended Audience

- Development team implementing the mobile app and backend services
- Municipal IT department evaluating the system for deployment
- Quality assurance team designing test cases
- Course instructor / evaluator assessing the requirements artifact

1.4 Definitions, Acronyms, and Abbreviations

- EXIF – Exchangeable Image File Format; metadata embedded in photos, including GPS coordinates
- GIS – Geographic Information System
- FR / NFR – Functional Requirement / Nonfunctional Requirement
- Ticket – A trackable work item created from a citizen's damage report

1.5 References

- PES University Dept. of CSE – Lab 1: Requirements Engineering & UML Use-Case Modelling (handout)
- Problem Statement #29 – Smart Cities, Transport & Logistics

2. Overall Description

2.1 Product Perspective

The app is a new, standalone citizen-engagement system that integrates with the municipality's existing GIS mapping platform and a third-party SMS/email/push notification gateway. It is not a replacement for internal municipal work-order systems but acts as the intake and routing layer in front of them.

2.2 Product Functions (Summary)

- Citizen submission of categorized, geotagged damage reports
- Automatic GPS extraction and GIS location pinning
- Automated ticket generation and department-based dispatch
- Engineer review, status update, and resolution workflow
- Citizen-facing status tracking and notifications
- Offline drafting with auto-sync when connectivity resumes

2.3 User Classes and Characteristics

- Citizen Reporter – general public, mobile-first users, variable technical literacy and network connectivity

- Municipal Engineer – municipal staff responsible for triaging and resolving tickets, access via web portal

2.4 Operating Environment

Citizen-facing client: Android and iOS mobile devices with camera and GPS. Engineer-facing client: desktop/laptop web browser. Backend: cloud-hosted application and database tier, integrated with the municipal GIS system and an external notification gateway (see Architectural Diagram).

2.5 Design and Implementation Constraints

- Must support low-connectivity areas via offline drafting and sync (NFR-001)
- Must comply with municipal data-privacy policy for citizen personal and location data (NFR-002)
- Must integrate with the existing municipal GIS platform's API

2.6 Assumptions and Dependencies

- Citizens' devices have a functioning camera and, where available, GPS/location services enabled
- The municipal GIS mapping system exposes an API or service the app can call to pin locations
- Department-to-category routing rules are maintained by municipal administrators and kept up to date

3. Specific Requirements

3.1 Functional and Nonfunctional Requirements

Full descriptions, measurable acceptance criteria, and rationale for each requirement are provided in the companion Requirements Table (Requirements_Table.docx). A condensed summary is provided below.

Req ID	Description	Priority
FR-001	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map.	High
FR-002	The system shall allow a Citizen Reporter to submit a damage report by selecting a category, entering a description, and attaching at least one photo.	High
FR-003	The system shall automatically dispatch a generated ticket to the relevant municipal engineering department based on the damage category.	High
FR-004	The system shall allow a Municipal Engineer to update ticket status (Received, In Progress, Resolved) with a resolution photo.	Medium
FR-005	The system shall allow a Citizen Reporter to track report status by Ticket ID and receive status-change notifications.	Medium
NFR-001	The reporting portal shall support offline report drafting and auto-sync when connectivity resumes.	High
NFR-002	The system shall encrypt all photo and GPS data in transit (TLS 1.2+) and at rest (AES-256), restricting access to authenticated staff.	High

3.2 Use Case Summary

The complete UML use-case diagram and detailed flow specification are provided in UseCase_Diagram.pdf and UseCase_Flow_Specification.docx. A summary of actors, use cases, and «include»/«extend» relationships is provided below.

UC ID	Use Case	Primary Actor	Relationship
UC-01	Submit Damage Report	Citizen Reporter	includes UC-02
UC-02	Upload Geotagged Photo	Citizen Reporter	includes UC-03
UC-03	Extract GPS Location	GIS Mapping System	—
UC-04	Track Report Status	Citizen Reporter	—
UC-05	Review & Dispatch Ticket	Municipal Engineer	extended by UC-07
UC-06	Update Repair Status	Municipal Engineer	extended by UC-08
UC-07	Escalate Priority Report	Municipal Engineer	extends UC-05
UC-08	Send Notification	Notification Service	extends UC-06

3.3 External Interface Requirements

- Municipal GIS Mapping System – receives extracted GPS coordinates and returns a map-pinned location reference for each report
- SMS / Email / Push Notification Gateway – receives status-change events and delivers notifications to the Citizen Reporter

4. Appendix

4.1 Related Artifacts

- Requirements_Table.docx – full FR/NFR table with acceptance criteria and rationale
- UseCase_Diagram.pdf – UML use-case diagram
- UseCase_Flow_Specification.docx – main success scenario and alternate flows for UC-01
- Architectural_Diagram.pdf – layered system architecture
- Work_Breakdown_Structure.docx – project phases, tasks, and estimated durations